

Latent Safe Update Derivation

Let z_{k+1}^{nom} denote the *nominal next latent state*, obtained directly from the current gradient-based approach.

We want to find a new next latent state z_{k+1}^{safe} such that:

$$z_{k+1}^{\text{safe}} = z_{k+1}^{\text{nom}} + \lambda d \quad (\text{A})$$

where d is the direction and λ is the step size along direction d .

This z_{k+1}^{safe} should satisfy:

$$B(z_{k+1}^{\text{safe}}) = (1 - \alpha\Delta) B(z_k) \quad (1)$$

Alternatively, this can be posed as an optimization problem:

$$z^{\text{safe}} = \arg \min_z \|z - z^{\text{nom}}\|^2 \quad \text{s.t. (1) holds.} \quad (2)$$

This can be solved if the direction d in (A) is chosen as:

$$d = \nabla B(z^{\text{nom}}) \quad (3)$$

Thus,

$$z_{k+1}^{\text{safe}} = z_{k+1}^{\text{nom}} + \lambda \nabla B(z_{k+1}^{\text{nom}}) \quad (4)$$

Now, we need to solve for the scalar λ .

A convenient way is to use a linear approximation:

$$B(z_{k+1}^{\text{safe}}) = B(z_{k+1}^{\text{nom}} + \lambda d) \quad (5)$$

$$\approx B(z_{k+1}^{\text{nom}}) + \nabla B(z_{k+1}^{\text{nom}})^\top \cdot \lambda d \quad (6)$$

Since

$$d = \nabla B(z_{k+1}^{\text{nom}}), \quad (7)$$

we obtain:

$$B(z_{k+1}^{\text{safe}}) \approx B(z_{k+1}^{\text{nom}}) + \lambda \|d\|^2 \quad (8)$$

Using constraint (1):

$$\lambda_{\text{linear}} = \frac{(1 - \alpha\Delta) B(z_k) - B(z_{k+1}^{\text{nom}})}{\|d\|^2} \quad (9)$$

Finally, this must satisfy:

$$\lambda_{\text{linear}} \geq 0 \quad (10)$$